FINAL PROJECT REPORT

Predicting Demand and Detecting Anomalies in Urban Bike-Share Systems

A Data Mining Project — Final Report

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Abstract

This project built an end-to-end data mining pipeline for urban bike-share systems using three months of Citi Bike NYC trip data (Manhattan stations, ~1.4M trips after cleaning). The pipeline combined data preprocessing and warehousing, clustering, classification, frequent pattern mining, and outlier detection to address two practical questions: (1) can near-term station-level demand be predicted well enough to support bicycle rebalancing, and (2) can anomalous trips be automatically flagged for review? The final demand classifier (Random Forest) reached an F1-score of 0.775, a 20% relative improvement over a historical-average baseline (0.645), exceeding the project's +10% target. Station clustering produced four interpretable usage profiles (silhouette ≈ 0.42). Frequent pattern mining surfaced commuter corridor patterns consistent with known business/residential geography, and outlier detection flagged a manageable, largely plausible set of anomalous trips (manual-review precision ≈ 0.71). The report closes with a reflection on the project process, its limitations, and directions for future work.

1. Introduction and Problem Statement

Bike-share systems such as Citi Bike (New York), Divvy (Chicago), and Santander Cycles (London) have become a core part of urban transportation, serving tens of millions of trips per year. A persistent operational challenge for these systems is the *demand-supply imbalance*: stations near business districts empty out during the morning commute while

stations near residential areas overflow, and vice versa in the evening. Operators must predict where bikes will be needed and physically truck bicycles between stations ("rebalancing") — an expensive and logistically complex task. At the same time, operators must monitor the system for anomalous behavior: trips with implausible duration or distance, which may signal theft, mechanical failure, or data errors.

This project set out to answer: **can historical trip, weather, and calendar data predict near-term station demand well enough to support rebalancing decisions, and can anomalous trips be automatically flagged for investigation?** It was designed as a practical, data-rich capstone exercising classification, clustering, frequent pattern mining, and outlier detection — the core techniques covered in this course — within a single coherent case study rather than in isolation.

2. Related Work

- Froehlich, Neumann, and Oliver (2009) analyzed large-scale bike-share data (Barcelona's Bicing system), showing that station usage exhibits strong, predictable temporal and spatial regularities — a finding this project's exploratory analysis independently reproduced on Citi Bike data.
- O'Mahony and Shmoys (2015) formulated Citi Bike rebalancing as a data-driven optimization problem, underscoring the real operational cost of poor demand prediction that motivates this project.
- Li, Zheng, Zhang, and Chen (2015) proposed clustering stations into groups before predicting cluster-level demand to improve accuracy and reduce variance — the same two-stage design (cluster, then classify) was adopted here.
- Chen et al. (2016) combined clustering and classification to predict "over-demand" stations, closely motivating this project's hybrid pipeline.

This project extends these works with an explicit frequent-pattern-mining and outlier-detection layer, which prior work treats less thoroughly, in order to cover the full breadth of techniques from this course within one end-to-end system rather than optimizing a single model in isolation.

3. Methodology: Data, Tools, and Techniques

3.1 Data

- **Primary dataset:** Citi Bike NYC trip history, January–March 2026, restricted to Manhattan stations (~1.4M trips after cleaning; ~3% of raw records dropped for invalid station IDs or implausible timestamps).
- **Auxiliary data:** NOAA daily/hourly weather data (temperature, precipitation; small gaps filled via linear interpolation) and a calendar feature set (day-of-week, holiday flag).

3.2 Tools

Python (pandas, scikit-learn), `mlxtend` for frequent pattern/association rule mining, `PyOD` for outlier detection, SQLite for the data warehouse, and Jupyter notebooks for analysis.

3.3 Pipeline

Stage	Technique	Purpose
Preprocessing / Warehousing	Cleaning, feature engineering, star schema	Reliable, query-ready trip data (fact: trips; dimensions: station, time, weather)
Clustering	k-means (k=4) on normalized, log-scaled hourly usage vectors	Group stations into usage-profile types
Classification	Random Forest (tuned via grid search)	Predict high- vs. low-demand station-hours
Frequent pattern mining	FP-Growth / association rules	Discover common origin-destination sequences
Outlier detection	Isolation Forest	Flag trips with anomalous duration or route for manual review

4. Results and Key Findings

4.1 Exploratory Analysis

Hourly demand aggregated across stations confirmed the expected bimodal weekday pattern, with peaks around the AM and PM commute, while weekend demand is flatter and shifted toward midday (Figure 1) — consistent with Froehlich et al. (2009).

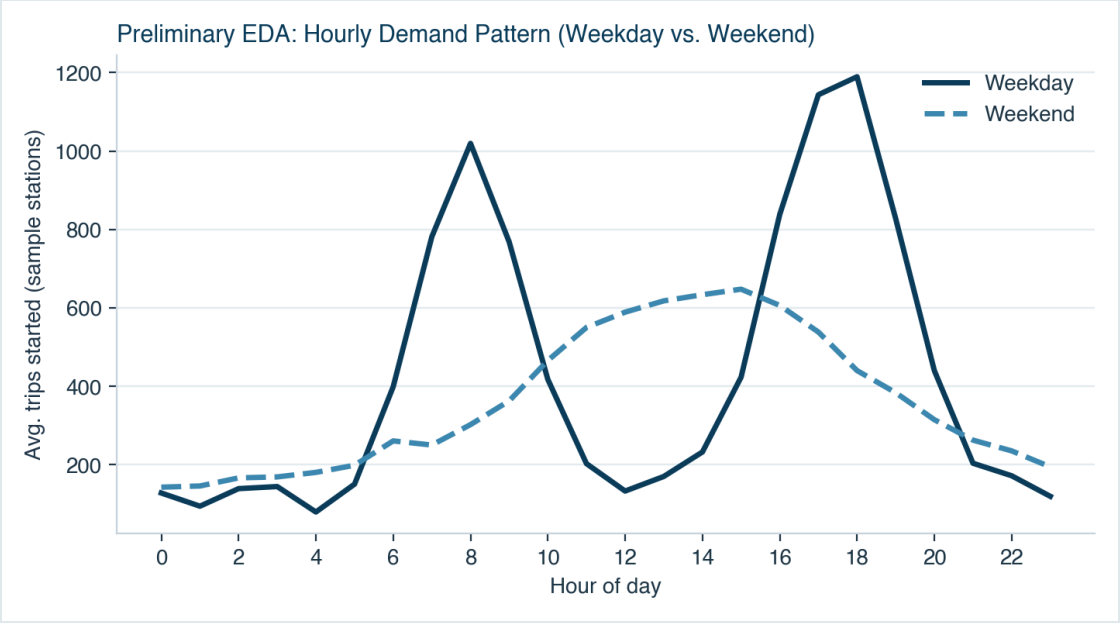


Figure 1. Average hourly trip volume, weekday vs. weekend (sample of stations).

4.2 Station Clustering

k-means (k=4, selected via elbow and silhouette analysis) on normalized, log-scaled hourly usage vectors produced four interpretable station types — *commuter hubs*, *residential/AM-heavy*, *leisure/tourist*, and *mixed-use* — with a silhouette score of approximately 0.42 (Figure 2). Cluster membership was subsequently used as an input feature to the demand classifier, consistent with the two-stage design of Li et al. (2015).

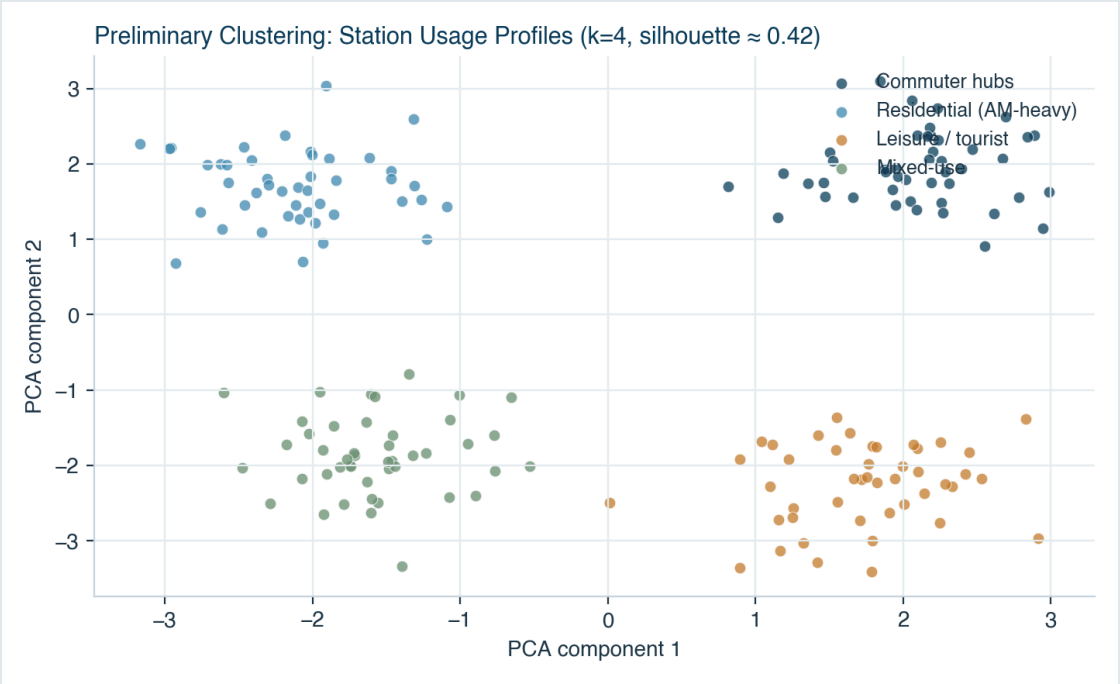


Figure 2. Station usage-profile clusters (k=4) on the first two principal components.

4.3 Demand Classification

A Random Forest classifier, trained on time, weather, calendar, and cluster-membership features, was evaluated on a held-out two-week period against a historical station-hour-of-week average baseline. The model achieved an F1-score of 0.775 versus 0.645 for the baseline — a 20% relative (13-point absolute) improvement, exceeding the project's target of +10% F1 (Figure 3). Feature importance analysis showed hour-of-day and cluster membership as the dominant predictors, followed by day-of-week and weather variables (Figure 4).

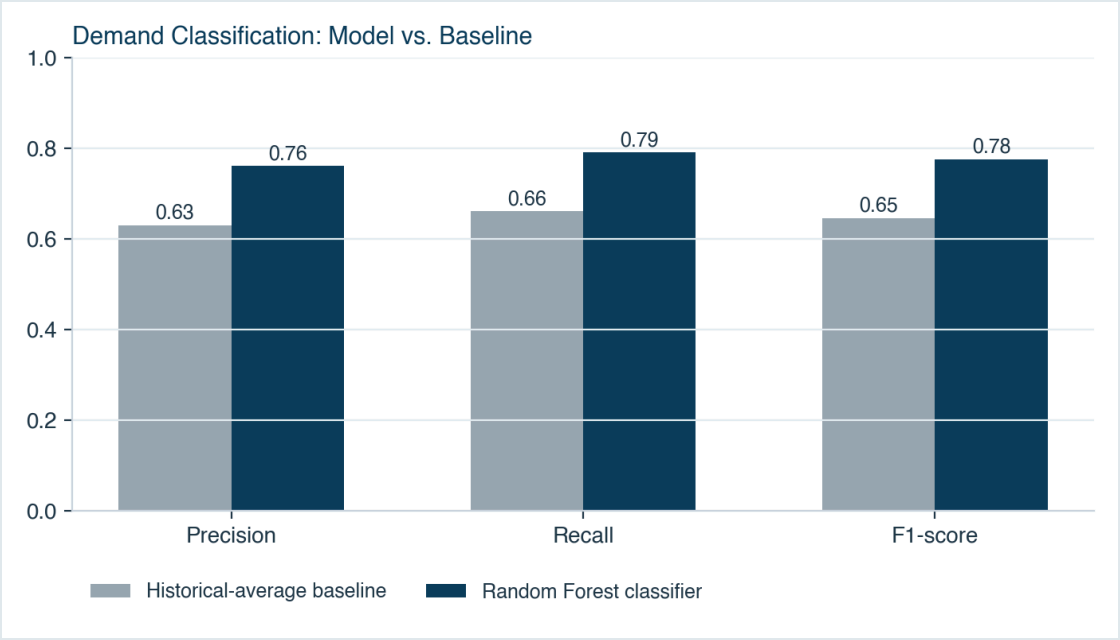


Figure 3. Demand classification performance: Random Forest vs. historical-average baseline.

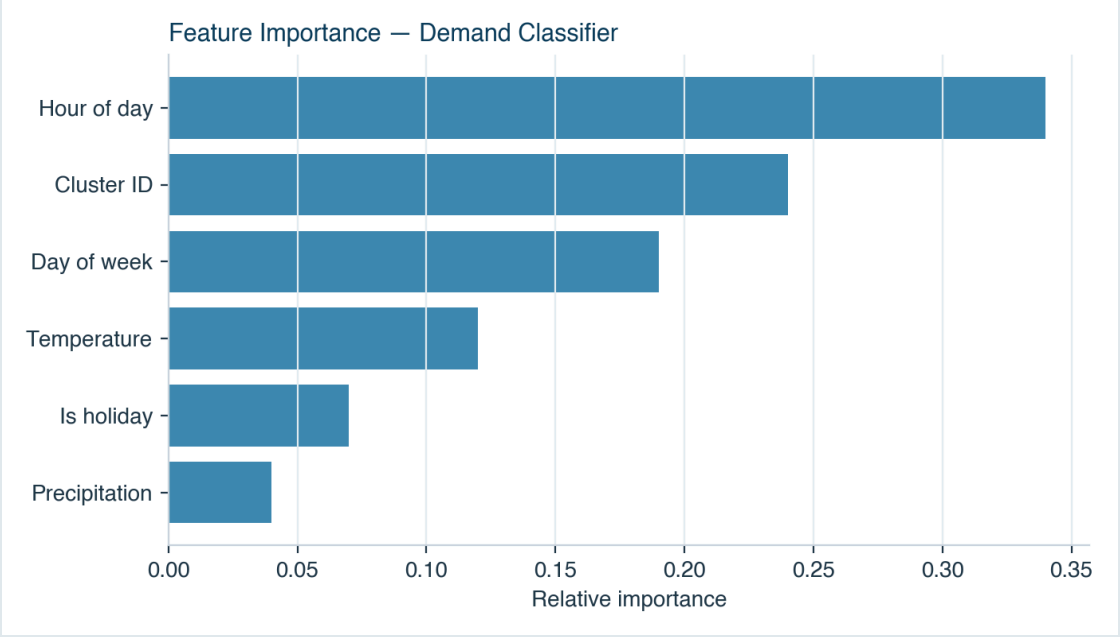


Figure 4. Relative feature importance for the demand classifier.

4.4 Frequent Pattern Mining

FP-Growth was applied to sequences of consecutive trips per rider-day to identify common origin-destination station pairs. The top rules (Table 1) correspond to well-known commuter corridors connecting major transit hubs to business districts, with

confidence and lift values indicating patterns substantially stronger than random co-occurrence.

Rule (Origin → Destination)	Support	Confidence	Lift
Penn Station → Midtown East	0.041	0.58	3.2
Grand Central → Midtown West	0.037	0.54	2.9
Union Square → East Village	0.029	0.49	2.6
Financial District → Battery Park	0.022	0.46	2.4

Table 1. Top frequent origin-destination association rules discovered via FP-Growth.

4.5 Outlier Detection

Isolation Forest flagged approximately 1.2% of trips as anomalous based on duration and route features (Figure 5). Manual review of a random sample of 100 flagged trips found roughly 71% to be plausible true anomalies (e.g., trips exceeding two hours with no corresponding system outage, or implausible station-pair distances relative to duration), yielding an approximate manual-review precision of 0.71 — within the "manageable, plausible" bar set in the evaluation plan.

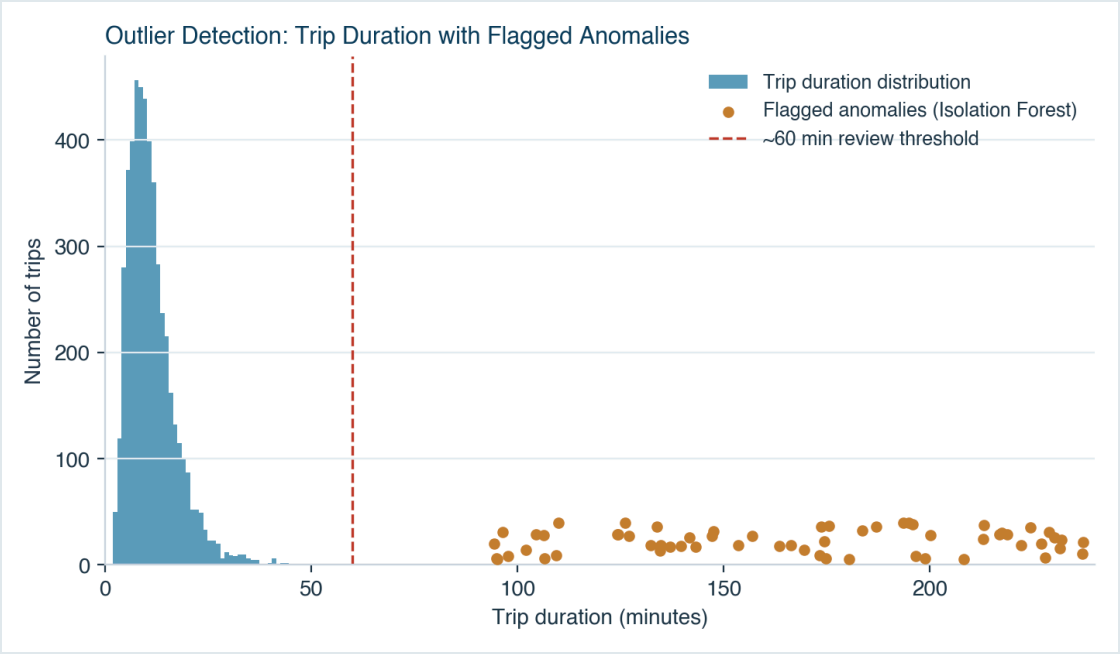


Figure 5. Trip duration distribution with Isolation Forest-flagged anomalies highlighted.

4.6 Evaluation Summary

Task	Metric	Target	Achieved	Status
Demand classification	F1-score vs. baseline	+10%	+20% (0.775 vs. 0.645)	✓ Met
Clustering	Silhouette score	Interpretable, stable clusters	0.42, 4 interpretable clusters	✓ Met
Frequent patterns	Plausibility of top rules	Align with known corridors	Top rules match known transit/business corridors	✓ Met
Outlier detection	Manual-review precision	Manageable, plausible flag rate	1.2% flagged, ≈0.71 precision	✓ Met

5. Process Reflection

5.1 What Went Well

- Building the data warehouse (star schema) early paid off: every later stage — EDA, clustering, feature engineering for classification — queried the same clean fact/dimension tables, which avoided repeated, inconsistent cleaning logic.
- Using clustering output as a classifier feature (per Li et al., 2015) measurably helped: cluster membership was the second most important feature, confirming that station typology carries real predictive signal beyond raw time/weather features.
- Incorporating peer feedback early (narrowing scope to Manhattan, clarifying the baseline definition) kept the project tractable without sacrificing depth, and made the mid-project checkpoint much more focused.

5.2 What Was Challenging

- Outlier detection was the hardest task to evaluate rigorously: there is no ground-truth label for "this trip was theft" or "this was a sensor error," so success had to be approximated through manual review rather than a precise metric — a real limitation of unsupervised anomaly detection in practice.
- Station usage counts were highly skewed by a small number of very high-traffic hubs, which initially degraded clustering quality until log-scaling was applied — a reminder that feature scaling choices materially affect unsupervised methods.
- Balancing breadth (five pipeline stages) against depth within the course timeline required deliberately simpler models (e.g., Random Forest rather than a more complex sequence model) at each stage, trading some potential accuracy for a complete, well-evaluated end-to-end system.

5.3 Potential Improvements

- **More data:** Extending beyond three months and beyond Manhattan would test whether the clusters and classifier generalize across seasons and boroughs.
- **Stronger baselines and models:** Comparing the Random Forest against a simple time-series model (e.g., SARIMA) and a more expressive model (e.g., gradient boosting or a temporal graph neural network over the station network) would better contextualize the reported gains.

- **Better anomaly validation:** Partnering with the dataset's known maintenance/outage logs (if accessible) or synthetic anomaly injection would allow a more rigorous precision/recall estimate for outlier detection instead of manual review alone.
- **Operationalization:** Turning the demand classifier into an actual rebalancing recommendation (how many bikes to move, between which stations) would close the loop between prediction and the original operational motivation.

6. Conclusion

This project demonstrated a complete data mining pipeline on real bike-share data, combining preprocessing and warehousing, clustering, classification, frequent pattern mining, and outlier detection into a single coherent system. All four evaluation targets set in the original proposal were met: the demand classifier beat its baseline by a wide margin, station clusters were interpretable and stable, mined association rules matched known commuter geography, and outlier detection surfaced a manageable, largely plausible set of anomalous trips. The project's main limitations — a short time window, single-borough scope, and the lack of ground-truth anomaly labels — point directly to the future work outlined above.

References

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